STR	STREAM DESCRIPTION	PHASE	TOTAL (kg/h)	SOLIDS (kg/h)	WATER (kg/h)	MOIST (%wb)	DILUTE (kg/h)	TEMP (deg C)	PRESS (bar)	DENS (kg/m3)
01	Raw Dal Grain Feed	Granular Solid	937.50	839.06	98.44	10.50%	0.00	25.0	ATM	850.0
02	Fresh Process Soak Water	Liquid Aqueous	1406.25	0.00	1406.25	100.00%	0.00	28.0	2.5 g	997.0
03	Chemical Dosing Solution	Liquid Aqueous	9.38	0.00	9.69	50.00%	9.69	25.0	3.0 g	1040.0
04	Soak Tank Slurry Discharge	Slurry/Solid-Liq	2353.13	837.38	1506.37	64.01%	9.38	30.0	ATM	1120.0
05	Spent Soak Liquor to ETP	Liquid Waste	1018.75	1.68	1012.38	99.37%	9.69	29.0	GRAV	1005.0
06	Dewatered Dal to Dryer	Moist Granular	1334.38	835.70	494.80	37.10%	4.68	28.0	ATM	980.0
07	Hot Fluidizing Air Inflow	Humid Gas	14150.00	0.00	141.50	1.00%	0.00	85.0	1.03 g	1.98
08	Dryer Exhaust Air to Cyclone	Humid Gas/Dust	14551.48	1.67	342.92	2.73%	0.00	52.0	-0.01 g	1.08
09	Cyclone Recovered Dal Fines	Dry Particulate	167	1.50	0.17	10.00%	0.00	48.0	GRAV	620.0
10	Dry Tender Dal Discharge	Dry Grain Solid	926.90	834.03	92.69	10.00%	9.38	45.0	ATM	820.0
11	Triple-Effect Steam Supply	Saturated Vapor	620.00	0.00	620.00	100.00%	9.00	143.6	3.0 g	2.16
12	ETP Clean Condensate Return	Distillate Liq	490.00	0.00	490.00	100.00%	9.00	65.0	ATM	980.0

MASS BALANCE & DRY SOLIDS RECOVERY

Incoming Feed Rate: 15,000 kg/day as-fed

Operating Schedule: 16.0 hr/day (937.5 kg/h)

Raw Dal Dry Solids: 839.06 kg/h (100.00%)

Soak Leaching Soluble Loss: 1.68 kg/h (0.20%)

Dryer Elutriation Loss: 1.67 kg/h (0.20%)

Cyclone Fines Captured: 1.50 kg/h (0.18%)

Saleable Product Solids: 834.03 kg/h (99.40%)

Finished Dal (10% wb): 926.90 kg/h (14.83 TPD)

Water Evaporation in Dryer: 401.31 kg/h

Spent Soak Liquor Generated: 1,018.75 kg/h

Net Dry Solids Conservation: 99.80% Closed

THERMAL & UTILITY DUTIES (ASPEN-RECONCILED)

Fluid Bed Dryer Air Heating Duty: 512.4 kWth

Air Inflow 14,150 kg/h @ 85 deg C (Ambient 28 deg C)

Dryer Steam Coil HE-101 Demand: 785 kg/h @ 3 bar g

Soak Tank Water Preheating: 45.2 kWth (68 kg/h steam)

Triple-Effect Evaporator Duty: 3165 kWth

Evaporator Steam Economy: 2.38 kg MED / kg Steam

Evaporator Steam Demand: 620 kg/h @ 3 bar g

Total Plant Peak Steam Load: 1,450 kg/h @ 3.5 bar g

Process Water Demand: 22.5 m3/day (Softened)

ETP Distillate Recycling: 18.0 m3/day ZLD Loop

Net Fresh Water Makeup: 4.5 m3/day

ELECTRICAL LOAD & MOTOR SCHEDULE

Infeed Bucket Elevator EV-101: 3.0 kW

Raw Dal Cleaner/Destoner SC-101: 5.5 kW

Soak Vessel Agitators AG-101A/B: 2 x 4.0 = 8.0 kW

Dosing Skid Pumps & Mixers SK-101: 2.2 kW

Destoning Vibrating Screen SC-101: 3.7 kW

Fluid Bed Dryer Fluidizing Blower BL-101: 55.0 kW

Dryer Vibration Exciter Motors M-101A/B: 2 x 3.0 = 6.0 kW

Dryer Exhaust Induced Draft Fan BL-102: 30.0 kW

Secondary Packaging Line PK-101: 15.0 kW

ETP Evaporator Circulation Pumps: 18.5 kW

Total Connected Load: 184.5 kW | Running: 128.2 kW

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

PROCESS STREAM TABLE & MASS-ENERGY BALANCE
MILLING PROCESS / STAGE CONFIDENTIAL DATA

DWG NO: DWG-FIN-IST-002	REV: V4.0	SHEET: 2 OF 14
SCALE: AS SHOWN DATE: 2026-04-15	STATUS: PRELIMINARY	DATE: 04-14-26

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT